

Differential transcendence of the Hahn–Exton q -cosine and the arithmetic of its zeros

Vico Bonfioli*

August 16, 2026

Abstract

Let $G(q, z) = {}_1\phi_1(0; q; q^2, z) = \sum_{k \geq 0} (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$, the Hahn–Exton (third Jackson) q -cosine. We prove that for every $0 < q < 1$ the difference Galois group over $\mathbb{C}(z)$ of the second-order q -difference equation satisfied by G contains SL_2 , and that the equation admits no $\sigma\delta$ -integrability relation; via the parametrized difference Galois theory of Hardouin–Singer [16] and Dreyfus–Hardouin–Roques [15] these give a second route to the differential transcendence of $G(q, \cdot)$ over $\mathbb{C}(z)$. That transcendence is itself known, and in stronger form: it is a corollary of Adamczewski–Dreyfus–Hardouin [10], and the unsolvability of the associated Riccati equation is Nishioka’s [20]; what is offered here is the explicit Galois-theoretic computation, valid also at algebraic q . We then prove the connection constant $C_1 = \lim_{R \rightarrow \infty} G(q, -R)/\theta_{q^2}(R) = 1/(q; q)_\infty$ against the divergent theta solution, and deduce from it the regularized product of the positive zeros, $\prod_{k \geq 1} z_k(q) q^{2(k-1)} = (q; q^2)_\infty$. The product is a corollary of that constant; an earlier attribution of the constant to Morita’s connection formula [19] is withdrawn in §4, where the reasons are given. We then pose the arithmetic question that these results frame: the transcendence of the zeros $z_k(q)$ at algebraic q . That question specializes to the transcendence of a specific reflection-group growth constant β_2 (OEIS A396406), for which we prove an unconditional effective non-rationality bound.

Keywords. Hahn–Exton q -cosine; third Jackson q -Bessel function; differential transcendence; parametrized difference Galois theory; q -difference equation; zeros of q -special functions; regularized product; partition integrality; effective irrationality.

2020 Mathematics Subject Classification. Primary 11J81, 33D15; Secondary 12H10, 33D45, 39A13, 11P84.

1 Introduction

The three Jackson q -analogues of the Bessel function are among the most-studied special functions of q -series. The third goes back to Hahn [28] and, in the form now standard, to Exton [26]; the attribution and the modern normalization are those of Koelink–Swarttouw [21] and Koornwinder–Swarttouw [18]. It has the q -cosine and q -sine specializations

$$G(q, z) := {}_1\phi_1(0; q; q^2, z) = \sum_{k \geq 0} \frac{(-1)^k q^{k(k-1)}}{(q; q)_{2k}} z^k,$$

*Independent researcher. vicobonfioli@gmail.com. Code and verification data: <https://github.com/ElVec1o/pythagorean-reflection-sequence>.

which is, up to an explicit prefactor, $J_{-1/2}^{(3)}(\sqrt{z}/q; q^2)$; it is entire of order 0 in z , and its positive zeros $0 < z_1(q) < z_2(q) < \dots$ are real and simple [21]. The analytic theory of these zeros (orthogonality, interlacing, asymptotics) is classical. Their *arithmetic* is, to our knowledge, untouched: nothing is known about the transcendence of $z_k(q)$ at algebraic q , the q -analogue of Siegel's theorem that the zeros of the Bessel function J_ν are transcendental.

This paper establishes the function-theoretic groundwork for that arithmetic and states the resulting transcendence problem precisely. Our results are of three kinds.

Galois theory (§3). We compute enough of the difference Galois structure of the defining equation to conclude differential transcendence.

Theorem 1 (known; Adamczewski–Dreyfus–Hardouin). *For every $0 < q < 1$, the Hahn–Exton q -cosine $G(q, \cdot)$ satisfies no algebraic differential equation over $\mathbb{C}(z)$.*

This is *not* new, and the known result is stronger. Adamczewski, Dreyfus and Hardouin [10, Thm. 1.2, case Q] prove that a Puiseux-series solution of a linear q -difference equation over $K = \bigcup_j \mathbb{C}(x^{1/j})$, with q not a root of unity, is either in K or hypertranscendental over K ; $G(q, \cdot)$ is entire and not in K , so the conclusion follows with no hypothesis to verify, over a larger base field, and for all q that are not roots of unity. It is recorded here only because the route below is different and yields the explicit Galois-theoretic statements of §3, which [10] does not compute. The unsolvability of the associated Riccati equation for $J_\nu^{(3)}$, and the non-solvability of the Galois group, were obtained for all ν by Nishioka [20] for q transcendental over $\mathbb{Q}$; the only increment here is that algebraic q are included. The engine is an unconditional proof that the difference Galois group contains SL_2 (Theorem 5), obtained by ruling out rank-1 submodules of every symmetric power through a growth estimate on asymptotic-expansion coefficients, together with the absence of a $\sigma\delta$ -integrability relation (Proposition 6).

A modular zero product (§4). Writing $Q = q^2$,

Theorem 2. $\prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_\infty$, and $G(q, z)/\theta_Q(-z) \rightarrow 1/(q; q)_\infty$ along the negative axis, where $\theta_Q(-z) = \sum_n (-1)^n q^{n(n-1)} z^n$.

The regularized product of the zeros is thus an explicit modular value: transcendental at algebraic q by Nesterenko's theorem [22].

Arithmetic of the zeros (§5). These results frame a clean conjecture.

Conjecture 3 (Z). *For algebraic $q \in (0, 1)$, every positive zero $z_k(q)$ of $G(q, \cdot)$ is transcendental.*

Two remarks fix the shape of this conjecture. First, no blanket statement of this kind can hold: the full Jacobi theta $\theta(z; q) = (z; q)_\infty (q/z; q)_\infty$ has zeros exactly at $z = q^{-k}$, algebraic whenever q is, so any hypothesis must exclude functions with a product formula, and it is the failure of the triple product for G that carries the conjecture. This is the structural analogue of Siegel's exclusion of 2λ odd. Second, the classical model needs stating with care, because the present object sits in the excluded case. Siegel [9] proves that for rational λ with 2λ *not* an odd integer, every nonzero zero of J_λ is transcendental; here $2\lambda = -1$, which is exactly what that hypothesis removes. For $\lambda = -\frac{1}{2}$ one has $J_{-1/2}(x) \propto x^{-1/2} \cos x$, whose zeros are the odd multiples of $\pi/2$, and their transcendence is Lindemann's theorem [8], not Siegel's. So the classical analogue of the conjecture below is Lindemann's, and the q -deformation is what makes it hard again (see also Lorch–Muldoon [6]). We have found no result, and no conjecture, in the literature on the arithmetic nature of zeros of q -series; the q -Bessel zero literature is uniformly classified as special functions, not number theory. We record where Conjecture 3 stands relative to the q -analogue of the Siegel–Shidlovskii method.

The position is weaker than a margin: as §5 sets out, the theorem of [12] yields linear independence over a fixed field containing q and not transcendence over $\overline{\mathbb{Q}}$, so no choice of parameters would make it apply to Conjecture 3; what the double-Pochhammer denominator $(q; q)_{2k}$ controls is the bookkeeping ratio of §5, which is a factor 2 past a boundary that is itself already out of reach (Remark 23). We also record an unconditional partial result for the specific zero of number-theoretic interest.

The motivating instance is the growth constant $\beta_2 = 1.4916177871\dots$ of the geodesic growth series of a planar right-triangle reflection group (OEIS A396406), which by [14] equals $1/\sqrt{q^*}$ where q^* is the least positive root of $S(q) := G(q, 2q(1-q))$; equivalently $z^* := 2q^*(1-q^*) = z_1(q^*)$ is the smallest positive zero of $G(q^*, \cdot)$. Conjecture 3 implies β_2 transcendental. We prove unconditionally (Proposition 24) that q^* has no rational representation of denominator $\leq 10^{1044}$.

Every numerical claim below is regenerated by a script committed in the repository above, named at the point of use, and each is reported with the agreement actually attained rather than with the working precision. The scripts are `qcos_identities.py` (symbolic identities), `qcos_lattice.py` (the integer zero lattice, exact in $\mathbb{Z}[[q]]$), `qcos_analytic.py` (the high-precision certificates, each run at two working precisions), `qcos_ledger.py` (the Diophantine-ledger figures), `qcos_nonintegrable.py` (the exact linear algebra behind §3), together with the Rust tool `tools/qcos_rank` for the rank exclusions of Remark 18 and `beta2_effective_irrationality.py` for Proposition 24.

2 The q -difference module

Throughout $Q = q^2$, $\sigma f(z) = f(Qz)$, and $a(z) = q + 1 - qz$. From the ${}_1\phi_1$ recurrence, G satisfies

$$q G(z) - a(z) G(Qz) + G(Q^2 z) = 0, \quad L := q - a\sigma + \sigma^2, \quad (1)$$

a second-order q -difference equation with companion matrix $A(z) = \begin{pmatrix} 0 & 1 \\ -q & a \end{pmatrix}$, $\det A = q$, regular (Poincaré-type) at $z = 0$ and irregular at $z = \infty$ (Newton slopes $\{0, 1\}$). A second solution is $\tilde{G}(z) = z^{1/2} H(z)$ with $H(z) = \sum_{k \geq 0} (-1)^k q^{k^2} (1-q) z^k / (q; q)_{2k+1}$; the pair (G, H) realizes the Hahn–Exton q -cosine/ q -sine $J_{\pm 1/2}^{(3)}$. The Casoratian $\text{Cas}(G, \tilde{G})(z) := G(z) \tilde{G}(Qz) - G(Qz) \tilde{G}(z) = c z^{1/2}$ is nonzero, and the exact q -Wronskian identity

$$q G(z) H(Qz) - G(Qz) H(z) = q - 1 \quad (2)$$

holds; it is reproduced to the full working precision, at least 120 digits, at three (q, z) points and at two precisions by `qcos_analytic.py`. We write M for the associated rank-2 σ -module over $\mathbb{C}(z)$.

At $z = \infty$ the equation has a formal solution basis reflecting the two Newton slopes. With $\theta_Q(x) = \sum_{n \in \mathbb{Z}} Q^{n(n-1)/2} x^n$, these are

$$f_1(z) = \theta_Q(-z) \hat{g}(z), \quad \hat{g}(z) = \sum_{k \geq 0} b_k z^{-k}, \quad b_k = \frac{q^{k-k(k-1)/2}}{(q; q)_k}; \quad f_2(z) = \frac{h(z)}{\theta_Q(-z/q)}, \quad h(z) = \sum_{k \geq 0} c_k z^{-k},$$

where $\hat{g}$ is q -Gevrey divergent (its q -Borel transform is $1/(q^2/\xi; q)_\infty$, so f_1 is Borel–Laplace summable off one Stokes direction) and h is convergent ($c_k = O(Q^{k^2/2})$). The Newton polygon, the q -Gevrey classification and the q -Borel–Laplace summation used here are those of Ramis, Sauloy and Zhang [34]. The coefficients b_k , which drive the Galois argument below, satisfy $b_k \asymp q^{-k^2/2}$; both $\hat{g}$ and h are determined by (1) and $b_0 = c_0 = 1$. Full details of the resummation are in [14]; here only the asymptotic expansions are used.

3 The difference Galois group

Lemma 4 (irreducibility of the σ -module). *For $0 < q < 1$ the rank-2 σ -module of G is irreducible over $\mathbb{C}(z)$: the Riccati equation*

$$u(z)u(Qz) - a(z)u(z) + q = 0, \quad a = q + 1 - qz,$$

has no solution $u \in \mathbb{C}(z)$.

Proof. Suppose $u = P/R$ with $P, R \in \mathbb{C}[z]$ coprime, of degrees p, r , leading coefficients α, β . Clearing denominators,

$$F(z) := P(Qz)P(z) - a(z)P(z)R(Qz) + qR(z)R(Qz) \equiv 0.$$

At $z = 0$: $F(0) = P(0)^2 - (q+1)P(0)R(0) + qR(0)^2 = 0$; if $R(0) = 0$ then $P(0) = 0$, contradicting coprimality, and symmetrically; so $P(0)R(0) \neq 0$ and $u(0) \in \{1, q\}$. *Top degree:* the three terms have degrees $2p, p+r+1, 2r$. If $p = r$, the top degree $2p+1$ is attained by the middle term alone, with coefficient $q\alpha\beta Q^r \neq 0$; if $|p-r| \geq 2$, the top degree is attained by the first or third term alone, with coefficient $\alpha^2 Q^p$ or $q\beta^2 Q^r$. Both are impossible, so $|p-r| = 1$. *Root pairing:* let $\zeta \neq 0$ be a root of R of multiplicity m , so $P(\zeta) \neq 0$. At $z = \zeta/Q$ the three terms of F have orders $k, k+m+[a(\zeta/Q)=0], m_0+m$, where $k = \text{ord}_{\zeta/Q} P$ and $m_0 = \text{ord}_{\zeta/Q} R$; since $F \equiv 0$ the minimum order cannot be attained by a single term, forcing $k \geq m_0 + m \geq m$. Symmetrically, a root $\xi \neq 0$ of P of multiplicity k forces $\text{ord}_{Q\xi} R \geq k$ (coprimality gives $\text{ord}_{Q\xi} P = 0$, so the first term has order k at ξ , the third $\text{ord}_{Q\xi} R$, and the middle at least $k + \text{ord}_{Q\xi} R$; again no single minimum). The two maps $\zeta \mapsto \zeta/Q, \xi \mapsto Q\xi$ are mutually inverse on the nonzero root sets, and chains close at length two by coprimality. Summing multiplicities, $p \leq r \leq p$, so $p = r$: contradicting $|p-r| = 1$. $\square$

Theorem 5 (the difference Galois group contains SL_2). *For every $0 < q < 1$, the difference Galois group over $\mathbb{C}(z)$ of the module of G contains SL_2 .*

Proof. For $N \geq 1$ suppose $\text{Sym}^N M$ has a rank-1 submodule, i.e. there is a nonzero solution w of the N -th symmetric-power equation with $r := w(Q \cdot)/w \in \mathbb{C}(z)$. The products $f_1^{N-k} f_2^k$ ($0 \leq k \leq N$) form a solution basis (their Casoratian is a power of $\text{Cas}(f_1, f_2) \neq 0$, by the standard symmetric-power formula), so $w = \sum_k c_k f_1^{N-k} f_2^k$ with c_k in the field of σ -constants; set $k_0 := \min\{k : c_k \neq 0\}$. In the analytic setting those σ -constants are the Q -elliptic functions rather than $\mathbb{C}$, which does not affect what follows: only $c_{k_0} \neq 0$ is used.

Case $k_0 < N$. Along a ray in a regular class, f_2/f_1 decays super-polynomially, so the rational function r has the Poincaré asymptotic expansion of the quotient of the dominant component:

$$r \sim (-1/z)^{N-k_0} (-z/q)^{k_0} \left(\frac{\hat{g}(Qz)}{\hat{g}(z)} \right)^{N-k_0} \left(\frac{h(Qz)}{h(z)} \right)^{k_0}.$$

The j -th coefficient of the formal series on the right is $(N-k_0)b_j Q^{-j}(1+O(q^{j-1}))$ with $b_j = q^{j-j(j-1)/2}/(q; q)_j$: the dominant term comes from $b_j Q^{-j}$ in $\hat{g}(Qz)/\hat{g}(z)$, and every competing product is smaller by at least q^{j-1} . That gap is not a numerical observation but an identity. For $1 \leq i \leq j-1$,

$$\frac{b_i b_{j-i}}{b_j} = q^{i(j-i)} \left[\begin{matrix} j \\ i \end{matrix} \right]_q, \quad (3)$$

since the powers of q contribute $\frac{1}{2}(j(j-1) - i(i-1) - (j-i)(j-i-1)) = i(j-i)$ and the Pochhammer factors assemble into the q -binomial coefficient, a polynomial in q with constant term 1. As $i(j-i) \geq j-1$ on that range, every split-off product carries at least one factor q^{j-1} relative

to b_j , uniformly in q and in N, k_0 . (An earlier version of this proof rested on a numerical check at a single j and a single q , and exhibited no c . The exponent identity, the polynomiality of $[i^j]_q$ and the bound $i(j-i) \geq j-1$ are now verified exactly over $\mathbb{Z}$ for $j \leq 399$ by `qcos_identities.py`, and machine-checked as `supermult_exponent` and `gap_ge`; see §7.) These coefficients grow like $q^{-j^2/2}$, whereas a rational function's expansion at ∞ has geometrically bounded coefficients. Since matching asymptotic expansions are unique, no rational r exists.

Case $k_0 = N$. Then $r = (-z/q)^N \rho^N$ with $\rho := h(Q \cdot)/h$ meromorphic on $\mathbb{C}^*$; $\rho^N \in \mathbb{C}(z)$ and single-valuedness force $\rho \in \mathbb{C}(z)$ (an N -th root of a rational function is single-valued on $\mathbb{C}^*$ only if every divisor multiplicity, including at 0 and ∞ , is divisible by N). Then $f_2(Qz)/f_2(z) = -(z/q)\rho(z)$ is rational, giving a rational solution of the rank-2 Riccati equation: contradicting Lemma 4.

Hence no $\text{Sym}^N M$ has a rank-1 submodule. $N = 1$ is irreducibility; $N = 2$ excludes the imprimitive (dihedral) case; and a projectively finite irreducible subgroup (A_4, S_4, A_5) fixes a line in some Sym^N with $N \leq 12$ (the classical binary polyhedral invariants), which is likewise excluded. By the classification of algebraic subgroups of GL_2 , as used in difference Galois theory in van der Put–Singer [35, Ch. 1], the group contains SL_2 . $\square$

The rank-two criterion used is that of Dreyfus, Hardouin and Roques [15, Prop. 2.9]; the corresponding dichotomy for general second-order linear difference equations, including the elliptic case, is Arreche, Dreyfus and Roques [13], which is later and does not record [15].

Proposition 6 (no $\sigma\delta$ -integrability). *For every $0 < q < 1$ there is no $B \in \text{gl}_2(\mathbb{C}(z))$ and no $c \in \mathbb{C}(z)$ with $\theta(A) = \sigma(B)A - AB + cA$, where $A = \begin{pmatrix} 0 & 1 \\ -q & a \end{pmatrix}$ and $\theta = z\partial_z$.*

Proof. Writing $B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, the entries give $\gamma = -q\beta_1$, $\delta = \alpha_1 + a\beta_1 - c$, and eliminating α (the central terms c cancel identically; verified symbolically in `qcos_nonintegrable.py`) leaves the necessary condition

$$qa\beta_3 = a_1(aa_1 - q)\beta_2 - a(aa_1 - q)\beta_1 + qa_1\beta + qz(a_1 + q^2a),$$

$\beta_k = \beta(Q^k z)$, $a = a(z)$, $a_1 = a(Qz)$. A polynomial part of degree $m \geq 0$ contributes top coefficient $\propto \beta_m(1 - q^{2+2m}) \neq 0$ at degree $m + 3$, above the inhomogeneity: so β has no polynomial part. If β has a pole off the orbit $z_a Q^{\mathbb{Z}}$ and away from $z = 0$, the maximal-modulus pole appears in exactly one term (the slot $z = \zeta Q^{-3}$, coefficient $qa \neq 0$ there), and the minimal-modulus pole likewise (slot $z = \zeta'$, coefficient qa_1): both impossible. The origin must be treated separately, since at $\zeta = 0$ one has $\zeta Q^{-3} = 0$ as well and all four β -terms are singular at once, so the isolation argument does not apply. Suppose then $\beta \sim Bz^{-n}$ at 0 with $B \neq 0$ and $n \geq 1$. Collecting the coefficient of z^{-n} in the displayed relation gives

$$(q+1)Bq(x-1)(x-q)(x-1/q), \quad x := q^{-2n},$$

which is nonzero for every $n \geq 1$ because $x = q^{-2n} > 1/q > 1 > q$; so β is regular at the origin. With that case closed, so the poles lie in the window $\{z_a Q^{-1}, \dots, z_a Q^3\}$ ($a(z_a Q^k) = (q+1)(1 - Q^k)$ vanishes only at $k = 0$). Sweeping the slots $z = z_a Q^{-4}, \dots, z_a Q^{-1}$ kills the orders at $z_a Q^{-1}, \dots, z_a Q^2$ one by one (each pole appears in a single term with nonvanishing coefficient), and the order at $z_a Q^3$ is killed at the slot $z = z_a Q$ (coefficient $a_1(aa_1 - q)$ with $(aa_1 - q)(z_a Q) = (q+1)^2(1 - q^2)(1 - q^4) - q$) or, where that vanishes (one root $q \approx 0.78027$), at the slot $z = z_a Q^2$: coefficient $a(aa_1 - q)$ with $(aa_1 - q)(z_a Q^2) = (q+1)^2(1 - q^4)(1 - q^6) - q$ (one root $q \approx 0.86753$); the two polynomials are coprime, so for every $q \in (0, 1)$ at least one slot applies. Hence $\beta \equiv 0$, contradicting $z(a_1 + q^2a) = z[(q+1)(1 + q^2) - 2q^3z] \neq 0$. (Cross-check, `qcos_nonintegrable.py`: the exact linear system for a β with poles of order ≤ 3 at the five window points has 15 unknowns and $\text{rank}(A) = 15$,

$\text{rank}[A|b] = 16$, hence no solution; the same holds at $q = \frac{1}{5}$. The two slot polynomials are located there as well, with roots $0.7802652115\dots$ and $0.8675323943\dots$ in $(0, 1)$, and shown to be coprime over $\mathbb{Q}[q]$. $\square$

Corollary 7 (differential transcendence of G). *For every $0 < q < 1$ the function $G(q, \cdot)$ satisfies no algebraic differential equation over $\mathbb{C}(z)$: the Hahn–Exton q -cosine is differentially transcendental.*

Proof, via parametrized difference-Galois theory. We apply the parametrized (σ, θ) -difference Galois theory with derivation $\theta = z\partial_z$. The determinant equation $\sigma(y) = \det(A)y = qy$ has the solution $y = z^{1/2}$ (indeed $\sigma(z^{1/2}) = (Qz)^{1/2} = qz^{1/2}$, as $Q = q^2$), which is algebraic over $\mathbb{C}(z)$; hence the parametrized difference Galois group of $\sigma(y) = \det(A)y$ lies in $\mathbb{C}^\times$. This is exactly the hypothesis of Dreyfus–Hardouin–Roques [15, Prop. 2.9], which for a rank-2 system whose difference Galois group contains SL_2 (here Theorem 5) gives the dichotomy: the parametrized group Gal^θ either is conjugate into $\text{GL}_2(\mathbb{C})$ (all solutions differentially algebraic) or contains $\text{SL}_2(\tilde{\mathbb{C}})$ (all nonzero solutions differentially transcendental), and the *first* alternative holds if and only if there exists $B \in \text{gl}_2(\mathbb{C}(z))$ with $\theta(A) = \sigma(B)A - AB$ (their integrability condition (7); $\theta(\det A) = 0$ since $\det A = q$ is constant in z , so no trace term appears). By Proposition 6 no such B exists: indeed none exists even allowing a scalar twist cA . Therefore the second alternative holds: $\text{Gal}^\theta \supseteq \text{SL}_2(\tilde{\mathbb{C}})$, and every nonzero solution, in particular $G(q, \cdot)$, is differentially transcendental over $\mathbb{C}(z)$. $\square$

4 The connection constant and the zero product

The behaviour of G at $z = \infty$ is governed by a theta-comparison. Let $\theta_Q(x) = \sum_{n \in \mathbb{Z}} Q^{n(n-1)/2} x^n$, so $\theta_Q(Qx) = x^{-1}\theta_Q(x)$ and, by the Jacobi triple product [30] (see [27, (1.6.1)]), $\theta_Q(-z) = (Q; Q)_\infty(z; Q)_\infty(Q/z; Q)_\infty$, an entire function of order 0 with simple zeros exactly at $z \in Q^\mathbb{Z}$. The following theorem evaluates the leading asymptotic constant of G and, as a consequence, the regularized product of its zeros; the constant $C_1 := \lim_{R \rightarrow \infty} G(-R)/\theta_Q(R)$ coincides with the Stokes-invariant connection coefficient of G against the theta solution.

Theorem 8 (the connection constant, and the zero product of the Hahn–Exton q -cosine). *For $0 < q < 1$,*

$$C_1 = \frac{1}{(q; q)_\infty}, \quad \text{and consequently} \quad \prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_\infty,$$

where $z_1 < z_2 < \dots$ are the positive zeros of $G(q, \cdot)$. Likewise, for the second solution's underlying entire function H with zeros $w_1 < w_2 < \dots$, $\prod_{k \geq 1} w_k(q) q Q^{k-1} = (q^3; q^2)_\infty$. (`qcos_analytic.py`: both products agree with the right-hand sides to the full working precision, 120 and 240 digits, at the three bases $q = 0.3$, $q = 0.42$ and $q = q^*$; the constant C_1 is approached along $z = uQ^{-N}$ at the rate $O(q^N)$, with $|(q; q)_\infty G/\theta_Q(-z) - 1| = 3.7 \cdot 10^{-48}$ at $q = 0.3$, $N = 45$.)

An earlier version of this note attributed the constant $C_1 = 1/(q; q)_\infty$ to Morita, as the specialisation to the q -cosine ($\nu = -\frac{1}{2}$) of his connection formula for $J_\nu^{(3)}$ [19, Thm. 1]. That attribution is withdrawn, on three counts. The base dictionary was wrong: Morita works with base p and $q = p^2$, so his p is the present q , not q^2 . His Theorem 1 carries the coefficients $1/(p^{\mp 2\nu}, p; p)_\infty$, and $(p^{-1}; p)_\infty = 0$, so the formula is *singular* at exactly $2\nu = \pm 1$, which is the q -cosine, the case wanted here. And he expands the solution at infinity in the basis at the origin, whereas C_1 is the coefficient against the divergent theta solution, which does not appear in his Theorem 1 at all. The constant is therefore proved here, not quoted, and the regularised zero product is a corollary of it rather than of [19]. Step (3) of the proof below re-derives the zero localisation $z_k = Q^{1-k}(1 + O(q^k))$ of Abreu–Bustoz–Cardoso [11, Thm. 2.3].

Proof. (1) An exact identity. From $(q; q)_{2k} = (q; q)_\infty / (q^{2k+1}; q)_\infty$ and Euler's expansion $(x; q)_\infty = \sum_{j \geq 0} (-1)^j q^{j(j-1)/2} x^j / (q; q)_j$ [27, (1.3.16)] at $x = q^{2k+1}$, together with $k(k-1) + 2kj = (k+j)(k+j-1) - j(j-1)$ and the substitution $m = k+j$, absolute convergence of the double sum permits the rearrangement

$$(q; q)_\infty G(z) = \sum_{m \geq 0} (-1)^m q^{m(m-1)} z^m P_m(1/z), \quad P_m(w) := \sum_{j=0}^m c_j w^j, \quad c_j = \frac{q^{j-j(j-1)/2}}{(q; q)_j}$$

(the c_j coincide with the b_j of §2; the identity is checked in $\mathbb{Q}[[q]]$, coefficient of z^k by coefficient of z^k , to order q^{90} or better for $k \leq 6$, and numerically to the full working precision at two precisions: `qcos_identities.py`).

(2) *The constant.* Fix $u < 0$ and put $z = uQ^{-N}$. The moduli $q^{m(m-1)}|z|^m$ peak at $m_0 = N + O(1)$ and decay like $q^{(m-m_0)^2 + O(1)}$ away from it. For $1 \leq j \leq m \leq 2N$, $2Nj - j(j-1)/2 \geq Nj$, so $|c_j z^{-j}| \leq Cq^{Nj}|u|^{-j}$ and $|P_m(1/z) - 1| \leq C'q^N$ uniformly; the terms with $m > 2N$ contribute a relative $O(q^{N^2})$. The negative- m part of $\theta_Q(-z)$ is $O(|z|^{-1})$, negligible against the peak. Hence $(q; q)_\infty G(z) / \theta_Q(-z) \rightarrow 1$ along $z = uQ^{-N}$, for each fixed $u < 0$.

(3) *Geometric zero dressing.* On the circle $|z - Q^{1-k}| = \varepsilon q^k Q^{1-k}$ the identity of step (1) and the bounds of step (2) give $|(q; q)_\infty G - \theta_Q(-z)| \leq Cq^k |\theta_Q(-z)|_{\text{peak}}$, while $\theta_Q(-z)$ has a simple zero at Q^{1-k} with derivative bounded below there; Rouché gives $z_k = Q^{1-k}(1 + O(q^k))$, i.e. $\log(z_k Q^{k-1}) = O(q^k)$, absolutely summable.

(4) *The product.* $G(z) = \prod_k (1 - z/z_k)$ (order 0, genus 0, $G(0) = 1$) and $\theta_Q(-z) = (Q; Q)_\infty (z; Q)_\infty (Q/z; Q)_\infty$ (Jacobi). Along $z = -R$, $R \rightarrow \infty$,

$$\frac{G(-R)}{\theta_Q(R)} = \frac{1}{(Q; Q)_\infty (-Q/R; Q)_\infty} \prod_{k \geq 1} \frac{1 + R/z_k}{1 + RQ^{k-1}},$$

and for $x, y > 0$ the function $R \mapsto \log \frac{1+Rx}{1+Ry}$ is monotone from 0 to $\log(x/y)$, so each factor is dominated by $|\log(z_k Q^{k-1})|$, summable by step (3). Dominated convergence gives $G(-R)/\theta_Q(R) \rightarrow 1/[(Q; Q)_\infty \prod_k z_k Q^{k-1}]$ as $R \rightarrow \infty$; evaluating along the subsequence $R = |u|Q^{-N}$ by step (2), the limit equals $1/(q; q)_\infty$, whence $C_1 = 1/(q; q)_\infty$ and $\prod_k z_k Q^{k-1} = (q; q)_\infty / (Q; Q)_\infty = (q; q^2)_\infty$ by Euler's $(q; q)_\infty = (q; q^2)_\infty (q^2; q^2)_\infty$. The H -statement is identical with comparison function $\theta_Q(-qz)$ and ratio $(1-q)/(q; q)_\infty$. $\square$

Remark 9 (the modular split at z^*). At $q = q^*$ the product splits off the candidate rational zero:

$$\prod_{k \geq 2} z_k(q^*) Q^{k-1} = \frac{(q; q^2)_\infty}{z^*} = 0.9882998846287462938546394 \dots$$

Were z^* (equivalently β_2) algebraic, the modular value $(q; q^2)_\infty$ (transcendental at algebraic q by Nesterenko's theorem [22] applied to the Euler products) would equal an algebraic number times the tail product $\prod_{k \geq 2} z_k Q^{k-1}$. This is consistent only if the tail product is itself transcendental; the arithmetic of the zero condition is thereby confined to that tail, which is the content of the multiplicity problem of [14].

5 The arithmetic of the zeros

5.1 Conjecture Z and its confluent evidence

Conjecture 3 is the q -analogue of the classical theorem (Siegel; see also Lindemann for the endpoint) that the positive zeros $j_{\nu, k}$ of the Bessel function J_ν are transcendental for rational ν . Two structural

facts support it. First, Theorem 1: G is differentially transcendental, so its zeros are not the zeros of any algebraic-differential shadow whose arithmetic could be controlled classically. Second, in the confluent limit $q \rightarrow 1$ the equation (1) degenerates to Bessel's equation, with the zeros rescaling as

$$\frac{z_k(q)}{(1-q)^2} \longrightarrow j_{-1/2,k}^2 \quad (q \rightarrow 1^-),$$

so Conjecture 3 deforms the classical statement. The scaling factor matters and an earlier version omitted it, writing $z_k(q) \rightarrow (qj_{-1/2,k})^2$; that is false, the zeros collapsing to 0 rather than to a finite limit. At $k = 1$ the measured ratios $z_1/(1-q)^2$ are 2.33902, 2.40449, 2.45502 at $q = 0.9, 0.95, 0.99$ against $j_{-1/2,1}^2 = (\pi/2)^2 = 2.46740$, whereas $z_1(0.99) = 2.45502 \cdot 10^{-4}$ against the discarded $(q\pi/2)^2 = 2.4183$. Here z_1 is located directly as the least positive zero of $G(q, \cdot)$ and not through the series of Theorem 10, which at these bases is outside its disc of convergence (`qcos_analytic.py`).

That confluence also explains, in one line, why the classical case is provable and the q -case is not. Write D_N for the denominator of the N -th partial sum. The classical engine (Lambert, Lindemann, and every factorial-denominator proof) runs on a *divergent step ratio*: $D_{N+1}/D_N = N+1 \rightarrow \infty$, so one further term is gained at asymptotically no arithmetic cost and $\text{tail} \cdot D_N \asymp 2/(N+1) \rightarrow 0$ (measured: 0.3097, 0.1629, 0.0991 at $N = 3, 6, 10$ for $e = \sum 1/k!$). The q -deformation replaces the factorial by $(q; q)_k$, whose step ratio is *constant* in the relevant scale: each further term gains $\asymp q^{2N}$ but costs $\asymp t^{2N}$, and at $s = 1$ the two cancel identically, so $\text{tail} \cdot D_N$ diverges (measured at $q = \frac{1}{3}$, $z_0 = 2q(1-q)$, with $D_N = t^{\deg_q P_N} = 3^{2N^2+N}$: 7.15, $7.47 \cdot 10^5$, $5.14 \cdot 10^{14}$ at $N = 2, 4, 6$; including the argument denominator 9^N only makes it worse). The bridge is explicit: $(q; q)_k / ((1-q)^k k!) \rightarrow 1$ as $q \rightarrow 1$ (measured at $k = 6$: 0.4719, 0.9277, 0.9925 at $q = 0.9, 0.99, 0.999$). All three tables are in `qcos_ledger.py`. The factorial (and with it the divergent step ratio that powers Lindemann) exists only in the confluent limit; at any $q < 1$ the ratio is constant, and a constant ratio can never win. This is the same exactness of Remark 23 seen dynamically, and it identifies precisely what a proof of Conjecture 3 must supply: an arithmetic mechanism with no classical shadow.

5.2 The self-referential specialization: β_2

The reflection-group growth constant β_2 is the reciprocal square root of the least positive root q^* of $S(q) = G(q, 2q(1-q))$, which is exactly the condition that the catalytic point $z^* = 2q^*(1-q^*)$ be a zero of $G(q^*, \cdot)$. Thus β_2 is governed by a *self-referential* specialization of Conjecture 3: the zero and the base move together along the diagonal $z = 2q(1-q)$.

At this point the q -Wronskian (2) yields a clean exact relation between two Hahn–Exton values: since $G(q^*, z^*) = 0$,

$$G(q^*, q^{*2}z^*) \cdot H(q^*, z^*) = 1 - q^* \quad (4)$$

(reproduced to the full working precision at two precisions, at least 120 digits, by `qcos_analytic.py`): a product of two q -Bessel values at the algebraic point z^* equal to the algebraic number $1 - q^*$. Together with $G(q^*, z^*) = 0$ and the modular zero product of Theorem 2, this places three exact constraints on the values at z^* ; a transcendence proof requires one more independent constraint than the reductio supplies, which is the precise sense in which Conjecture 3 is a multiplicity (rather than a single-relation) problem.

5.3 The q -Siegel–Shidlovskii margin

The natural route to Conjecture 3 is the q -analogue of the Siegel–Shidlovskii method developed by Amou–Matala-aho–Väänänen [12]. Two things must be said precisely, since the terminology is ours

and not theirs. Their theorem applies to systems of the shape $z^s y(qz) = a(z)Cy(z) + b(z)$ with C a *constant* matrix, not to Poincaré-type systems in general; and their hypothesis is arithmetic on q , namely (5.4) of [12] in terms of $\lambda = \log H(q)/\log \|q\|_v$, not a margin in our sense. What their theorem yields is *linear independence over the fixed field $\mathbb{K}$* containing q , never transcendence over $\overline{\mathbb{Q}}$; no q -analogue of Siegel–Shidlovskii giving transcendence is known. The quantity we call the margin below is a convenient bookkeeping ratio for the same obstruction: the ratio of the logarithmic height of the cleared denominators to that of the analytic decay. A reader working on the value side should compare Amou–Katsurada–Väänänen [1], who for a q -analogue of J_0 determine exactly which $\alpha \in \mathbb{K}^\times$ have value in $\mathbb{K}$; we make no value-level claim of that kind here, and the question addressed below concerns the *zeros*, on which we have found nothing in the literature. For the Hahn–Exton class this ratio is

$$\gamma = \frac{\log(t/s)}{2 \log t} \leq \frac{1}{2} \quad (q = s/t \text{ in lowest terms}),$$

the factor 2 being the double Pochhammer $(q; q)_{2k}$; this is below the threshold, so the method does not apply as stated. Extending it past the margin for this irregular-at- ∞ class is the outstanding problem; the connection data needed for the irregular direction is available in closed form [14].

5.4 Integrality of the zero lattice

The zeros, viewed as functions of the base, are integer power series after the Newton-polygon rescale: all of them, not only the first.

Theorem 10 (integrality of the zero lattice). *For every $k \geq 1$ the rescaled k -th zero $u_k(q) := q^{2(k-1)}z_k(q)$ lies in $1 + q\mathbb{Z}[[q]]$. In particular*

$$z_1(q) = 1 - q - q^3 + q^4 - q^5 + 2q^6 - 2q^7 + 4q^8 - 6q^9 + 8q^{10} - 14q^{11} + 21q^{12} - \dots \in \mathbb{Z}[[q]].$$

The *shape* of such an expansion is known: Ismail and Zhang [17] show that $q^{2n-1}z_n$ is analytic in q^n , with coefficients polynomial in three generators over rational functions of q , and the relevant entire function lies in the admissible class of Hayman [29]. What is proved here, and what we have not found in that literature, is that the coefficients are *integers*.

Proof. From $g_{k'} = (-1)^{k'} q^{k'(k'-1)} / (q; q)_{2k'}$ and the exponent identity $k'(k'-1) - 2(k-1)k' + k(k-1) = (k' - k)(k' - k + 1)$,

$$F_k(q, u) := q^{k(k-1)} G(q, u q^{-2(k-1)}) = \sum_{k' \geq 0} (-1)^{k'} q^{(k'-k)(k'-k+1)} \frac{u^{k'}}{(q; q)_{2k'}}.$$

The exponent $(k' - k)(k' - k + 1)$ is a product of consecutive integers, hence nonnegative, and vanishes exactly at $k' \in \{k-1, k\}$: each zero scale is carried by its own edge of the strictly convex Newton polygon of G . Since $1/(q; q)_{2k'} \in \mathbb{Z}[[q]]$ (partitions into parts $\leq 2k'$), $F_k \in \mathbb{Z}[[q]][[u]]$ with $F_k(0, u) = (-1)^{k-1} u^{k-1} (1 - u)$; at $u = 1$ the value vanishes and $\partial_u F_k(0, 1) = (-1)^k$ is a unit, so Hensel’s lemma in the complete local ring $\mathbb{Z}[[q]]$ yields a unique root $u_k \in 1 + q\mathbb{Z}[[q]]$. The identification of this formal root with the analytic k -th zero is not formal: it holds where the series converges, and u_1 has radius $|q_c| = 0.5546\dots$ by Remark 18(e). For $0 < q < |q_c|$ the scale localization $z_k = q^{2(1-k)}(1 + O(q^k))$ of §4 pins the root to the k -th zero; outside that disc the statement of the theorem is about the power series and not about the zero. (The lattice is regenerated exactly in $\mathbb{Z}[[q]]$ to order q^{300} for $k \leq 8$ by `qcos_lattice.py`: $u_2 = 1 - q^6 - 2q^8 + q^9 - \dots$, $u_3 = 1 - q^{15} - \dots$, $u_4 = 1 - q^{28} - \dots$. The onset law visible here is proved next. The Newton recursion is carried out over $\mathbb{Z}$ and its residual verified to order q^{80} for $k \leq 6$ in Lean; the Hensel step itself is not formalised, see §7.) $\square$

5.5 Lattice values and the hexagonal law

The deviation $u_k - 1$ begins exactly at the k -th hexagonal number $k(2k - 1)$. This reduces to, and is proved by, an exact evaluation of G at its own lattice points.

Lemma 11 (parameter–argument swap; Koornwinder–Swarttouw). *For $|Q| < 1$, $(b; Q)_\infty {}_1\phi_1(0; b; Q, z)$ is symmetric in (b, z) ; equivalently*

$$(q; Q)_\infty G(q, z) = \sum_{n \geq 0} \frac{(-1)^n q^{n^2}}{(Q; Q)_n} (zQ^n; Q)_\infty. \quad (5)$$

This is Koornwinder–Swarttouw [18, Prop. 2.1, eq. (2.3)], with the same double-sum proof; they note in their Remark 2.2 that it is a limit case of Heine’s transformation [27, §1.4]. It is restated here for use below, not claimed.

Proof. In $(b; Q)_\infty {}_1\phi_1(0; b; Q, z) = \sum_n (-1)^n Q^{\binom{n}{2}} z^n (bQ^n; Q)_\infty / (Q; Q)_n$ expand $(bQ^n; Q)_\infty = \sum_i (-1)^i Q^{\binom{i}{2}} (bQ^n)^i / (Q; Q)_i$ (Euler, [27, (1.3.16)]); since $\binom{n}{2} + \binom{i}{2} + ni = \binom{n+i}{2}$, the double sum is $\sum_{n,i} (-1)^{n+i} Q^{\binom{n+i}{2}} z^n b^i / ((Q; Q)_n (Q; Q)_i)$, manifestly symmetric. For (5) take $b = q$ and use $(z; Q)_\infty / (z; Q)_n = (zQ^n; Q)_\infty$ on the swapped side, with $Q^{\binom{n}{2}} q^n = q^{n^2}$. (`qcos_identities.py`: the symmetry holds to the full working precision at a generic (Q, b, z) at two precisions, and (5) itself is verified exactly in $\mathbb{Q}[[q]]$ to order q^{250} at three rational z .) $\square$

Proposition 12 (square vanishing at the lattice). *For every $m \geq 0$,*

$$G(q, Q^{-m}) = (-1)^{m+1} \frac{(Q; Q)_\infty}{(q; Q)_\infty} \sum_{r \geq 0} \frac{(-1)^r q^{(m+1+r)^2}}{(Q; Q)_r (Q; Q)_{m+1+r}},$$

so $\text{ord}_q G(q, Q^{-m}) = (m+1)^2$ exactly, with leading coefficient $(-1)^{m+1}$.

This follows in two lines from Koornwinder–Swarttouw [18, Prop. 2.1 and Rem. 2.3, eq. (2.6)]: putting $n = m+1$, $z = q$ in their (2.6) gives the leading factor $q^{(m+1)^2}$ and the r -th term exponent $(m+1+r)^2$, which is the display above. It is stated separately only because §5 uses it repeatedly.

Proof. Evaluate (5) at $z = Q^{-m}$. The factor $(Q^{n-m}; Q)_\infty$ vanishes for $n \leq m$ and equals $(Q; Q)_\infty / (Q; Q)_{n-m-1}$ for $n \geq m+1$; reindexing $n = m+1+r$ gives the display. The q -order of the r -th term is $(m+1+r)^2$, strictly increasing in r , and all other factors are units in $\mathbb{Z}[[q]]$, so no cancellation can occur and the $r = 0$ term decides. (`qcos_identities.py` verifies the identity in $\mathbb{Q}[[q]]$ to order q^{230} or better for $m \leq 4$, confirms $\text{ord}_q G(q, Q^{-m}) = (m+1)^2$ there, and reproduces it numerically at $q = 0.3$, $m = 3$ to 46 and 107 digits at working precisions 60 and 120.) $\square$

Corollary 13 (hexagonal law). *For every $k \geq 1$,*

$$u_k(q) = 1 - q^{k(2k-1)}(1 + O(q)) :$$

the deviation of the k -th rescaled zero from 1 begins exactly at the k -th hexagonal number, with leading coefficient -1 .

Proof. $F_k(q, 1) = q^{k(k-1)} G(q, Q^{1-k}) = (-1)^k q^{k(2k-1)}(1 + O(q))$ by Proposition 12 at $m = k-1$ (note $k(k-1) + k^2 = k(2k-1)$), while $\partial_u F_k(q, 1) = (-1)^k(1 + O(q))$. The first Newton correction from $u = 1$ is therefore $-q^{k(2k-1)}(1 + O(q))$, and all subsequent corrections are $O(q^{2k(2k-1)})$. This proves the onsets 1, 6, 15, 28 observed above and predicts $u_5 = 1 - q^{45} - \dots$; the onsets $k(2k-1)$ with leading coefficient -1 are confirmed for $k \leq 8$ in `qcos_lattice.py`. $\square$

Remark 14. Three readings of Proposition 12. (a) The r -sum is again a Hahn–Exton series, now of integer order $m+1$ in base Q : the q -cosine ($\nu = -\frac{1}{2}$) sampled on its own zero lattice reproduces the integer-order family: an order reflection $-\frac{1}{2} \mapsto m+1$. (b) Along any backward orbit $z = Q^{-m}w$, (5) gives $G(Q^{-m}w) \sim (Q^{-m}w; Q)_\infty / (q; Q)_\infty$ as $m \rightarrow \infty$, with constant $(w; Q)_\infty (Q/w; Q)_\infty / (q; Q)_\infty = \theta_Q(-w) / (q; q)_\infty$ by the triple product: recovering the connection constant of the backward normal form independently. (c) At any zero $z_k(q)$ of G , (5) transfers the vanishing into the *parameter* slot: ${}_1\phi_1(0; z_k(q); Q, q) = 0$. At the crossing $z_1(q^*) = 2q^*(1 - q^*)$ of the reductio this is a ${}_1\phi_1$ -vanishing with all-rational parameters; its truncation ledger has the same heights as the original (the swap preserves the exactness of Remark 23).

Proposition 15 (sine side: integrality, lattice values, onsets). *All of the above has a sine-side counterpart. (a) The zeros $w_k(q)$ of H satisfy $v_k := q^{2k-1}w_k(q) \in 1 + q\mathbb{Z}[[q]]$: with $(q^2; q)_{2k'} = (q; q)_{2k'+1} / (1 - q)$,*

$$q^{k(k-1)} H(q, v q^{1-2k}) = \sum_{k' \geq 0} (-1)^{k'} q^{(k'-k)(k'-k+1)} \frac{v^{k'}}{(q^2; q)_{2k'}},$$

the same strictly convex exponent as in Theorem 10, so Hensel applies on every edge. (b) Writing $H(q, z) = {}_1\phi_1(0; q^3; Q, qz)$, the dressing evaluates H on its own lattice: for $m \geq 0$,

$$H(q, q^{1-2m}) = (-1)^m \frac{(Q; Q)_\infty}{q (q^3; Q)_\infty} \sum_{r \geq 0} \frac{(-1)^r q^{(m+r+1)^2}}{(Q; Q)_r (Q; Q)_{m+r}},$$

so $\text{ord}_q H(q, q^{1-2m}) = (m+1)^2 - 1$. (c) Consequently $v_k = 1 - q^{k(2k+1)}(1 + O(q))$ for every $k \geq 1$.

Proof. (a) is the displayed identity together with the argument of Theorem 10: the exponent $(k' - k)(k' - k + 1)$ is strictly convex in k' with a unique minimiser, so the Newton polygon has a single edge of slope 0 and Hensel applies.

(b) Apply Lemma 11 with $b = q^3$ and $Q = q^2$ to $H(q, z) = {}_1\phi_1(0; q^3; Q, qz)$:

$$(q^3; Q)_\infty H(q, z) = \sum_{n \geq 0} (-1)^n q^{n^2+2n} \frac{(qzQ^n; Q)_\infty}{(Q; Q)_n}.$$

At $z = q^{1-2m}$ one has $qz = q^{2-2m} = Q^{1-m}$, so $qzQ^n = Q^{n+1-m}$ and the factor $(Q^{n+1-m}; Q)_\infty$ vanishes for $n \leq m-1$, the product then containing the factor $1 - Q^0$. For $n \geq m$ it equals $(Q; Q)_\infty / (Q; Q)_{n-m}$. Writing $n = m + r$ and using $(m+r)^2 + 2(m+r) = (m+r+1)^2 - 1$ turns the sum into the displayed one, the q^{-1} prefactor being the -1 in that exponent. The order statement follows since the $r = 0$ term is the unique one of minimal q -order.

(c) Combine (a) and (b): v_k is the Hensel root attached to the edge, and the first correction to $v_k = 1$ enters at the q -order of the rescaled function at $v = 1$, which by (b) at $m = k$ is $k(k-1) + \text{ord}_q H(q, q^{1-2k}) = k(k-1) + ((k+1)^2 - 1) = k(2k+1)$, exactly as on the cosine side with T_{2k-1} replaced by T_{2k} (identity `sine_onset` of §7). (`qcos_lattice.py` verifies integrality of v_k and the onsets $k(2k+1)$, with leading coefficient -1 , exactly in $\mathbb{Z}[[q]]$ to order q^{300} for $k \leq 6$.) $\square$

Remark 16 (the triangular law). $k(2k-1) = T_{2k-1}$ and $k(2k+1) = T_{2k}$ are consecutive triangular numbers, so the deviation onsets of the two interleaved zero lattices are

$$\text{onset}(u_k - 1) = T_{2k-1}, \quad \text{onset}(v_k - 1) = T_{2k},$$

with leading coefficient -1 in both families: between them, the cosine and sine lattices switch on at every triangular number exactly once, alternating between the families. (Verified exactly in $\mathbb{Z}[[q]]$ through u_8 , i.e. $T_{15} = 120$, and v_6 , i.e. $T_{12} = 78$; `qcos_lattice.py`.)

Proposition 17 (stable deviation law). *The two deviation series stabilize, q -adically as $k \rightarrow \infty$, to one and the same eta-quotient:*

$$(1 - u_k(q))q^{-T_{2k-1}} \longrightarrow \frac{1}{(q^2; q^2)_\infty^2} \longleftarrow (1 - v_k(q))q^{-T_{2k}},$$

whose coefficients $1, 2, 5, 10, 20, 36, 65, \dots$ count two-colored partitions. The agreement depth measured exactly in $\mathbb{Z}[[q]]$ is $2k - 2$ on the cosine side for $2 \leq k \leq 8$ and $2k - 1$ on the sine side for $2 \leq k \leq 6$ (`qcos_lattice.py`).

Proof. The convergence asserted is q -adic (coefficient-by-coefficient in $\mathbb{Z}[[q]]$), so it suffices to show each of three exact power-series limits below holds with agreement depth $\rightarrow \infty$ in k ; no analytic interchange is involved.

Newton correction. Since u_k is the Hensel root of $F_k(q, \cdot)$ at $u = 1$ with $F_k(q, 1) = O(q^{T_{2k-1}})$ (Corollary 13) and $\partial_u F_k(q, 1)$ a unit, one Newton step gives $1 - u_k = F_k(q, 1)/\partial_u F_k(q, 1) \cdot (1 + E_k)$ with $E_k = O(q^{T_{2k-1}})$ (the next correction is quadratic in $F_k(q, 1)$). As $T_{2k-1} = k(2k - 1) \rightarrow \infty$, $E_k \rightarrow 0$ q -adically, so it suffices to evaluate the ratio $F_k(q, 1)/\partial_u F_k(q, 1)$ scaled by $q^{-T_{2k-1}}$.

Numerator. By Proposition 12 at $m = k - 1$,

$$(-1)^k F_k(q, 1) q^{-T_{2k-1}} = \frac{(Q; Q)_\infty}{(q; Q)_\infty (Q; Q)_k} + O(q^{2k+1}) \xrightarrow{q\text{-adic}} \frac{1}{(q; Q)_\infty},$$

the r -th term of the lattice sum contributing, after the prefactor $q^{k(k-1)}$ and the rescaling by $q^{-T_{2k-1}}$, at order $q^{(k+r)^2 - k^2}$, so that $r \geq 1$ is $O(q^{2k+1})$, and $(Q; Q)_k = (Q; Q)_\infty (1 + O(q^{2k+2}))$. Agreement depth $2k + 1$.

Denominator. In $\partial_u F_k(q, 1) = \sum_{k' \geq 0} (-1)^{k'} k' q^{(k'-k)(k'-k+1)}/(q; q)_{2k'}$ the indices $k' = k + j$ and $k' = k - 1 - j$ share the exponent $(k' - k)(k' - k + 1) = j(j + 1)$ with opposite parity and weights $k + j, k - 1 - j$; their sum has weight $(k + j) - (k - 1 - j) = 2j + 1$, whence

$$(-1)^k \partial_u F_k(q, 1) = \frac{\sum_{j \geq 0} (-1)^j (2j + 1) q^{j(j+1)}}{(q; q)_\infty} + O(q^{2k-1}) \xrightarrow{q\text{-adic}} \frac{(q^2; q^2)_\infty^3}{(q; q)_\infty}$$

by Jacobi's identity $\sum_{j \geq 0} (-1)^j (2j + 1) q^{j(j+1)} = (q^2; q^2)_\infty^3$ [30]. Agreement depth $2k - 1$.

Quotient. Dividing and using the exact splitting $(q; q)_\infty = (q; Q)_\infty (Q; Q)_\infty$,

$$\frac{1/(q; Q)_\infty}{(q^2; q^2)_\infty^3 / (q; q)_\infty} = \frac{(q; q)_\infty}{(q; Q)_\infty (Q; Q)_\infty^3} = \frac{(Q; Q)_\infty}{(Q; Q)_\infty^3} = \frac{1}{(Q; Q)_\infty^2} = \frac{1}{(q^2; q^2)_\infty^2}.$$

All three limits are exact with depth growing in k and $E_k \rightarrow 0$, so the product limit is exact; this proves the cosine case. The sine case is identical with the numerator limit $1/(q^3; Q)_\infty$ (Proposition 15) and $(q^2; q)_\infty = (q; q)_\infty / (1 - q)$ in the denominator, the two $(1 - q)$ factors cancelling to give the same $1/(q^2; q^2)_\infty^2$. (The measured cosine agreement depth is exactly $2k - 2$ through $k = 8$ and the sine exactly $2k - 1$ through $k = 6$, consistent with the depth- $(2k - 1)$ bound above.) $\square$

Remark 18 (arithmetic and analytic features of the zero curve). (a) The series satisfies no algebraic relation $P(q, z_1) = 0$ of bidegree $\leq (6, 5)$; in particular it is not a rational function of numerator and denominator degrees within that bidegree. Nothing here excludes an algebraic relation of larger bidegree, and the statement is exactly the bounded-profile one. (Rank certificate: the 42 columns have full rank over the first 106 coefficients, at each of two primes; `tools/qcos_rank`.) (b) Its radius of convergence is $|q_c| = 0.5545\dots$, set by the real fold point of item (e); in particular $q^* = 0.4494\dots$ lies inside the disc of convergence. (c) The integrality opens a *linear-height* approximation ledger: at

$q = s/t$ the N -th partial sum is rational with denominator t^N and error $\asymp (q/R)^N$, $R = 0.5545\dots$, so the Liouville criterion for the crossing $z_1(q) = 2q(1-q)$ reads $s < R$: failing, but now only by the factor $s/R \approx 1.80$ at $s = 1$, the narrowest margin among all ledgers recorded for this problem; the obstruction has become the sub-unit radius, i.e. the complex collision of zeros. (d) The integer coefficient sequence $1, -1, 0, -1, 1, -1, 2, -2, 4, -6, 8, -14, 21, -34, 56, -88, 148, -242, 398, -669, 1109, \dots$ appears to be new.

(e) The radius-limiting singularity is located and characterized: it is the simple fold $q_c = -0.5545786101465797\dots$, $z_c = 2.5239525204417103\dots$ (both regenerated to 230 digits at two working precisions by `qcos_analytic.py`), where the analytically continued branch collides with a second zero branch on the negative q -axis ($G = G_z = 0$ with $G_{zz} = 0.2143\dots \neq 0$, $G_q = -1.5012\dots \neq 0$, so $z_1(q) = z_c \pm \tilde{C}\sqrt{q-q_c} + \dots$ with $\tilde{C}^2 = 2G_q/(-G_{zz})$ at the fold). The fold law now has a closed-form constant:

$$|c_n| n^{3/2} |q_c|^n \longrightarrow \frac{|\tilde{C}| \sqrt{|q_c|}}{2\sqrt{\pi}} = 0.7863083135\dots,$$

The constant is computed from the fold data at two working precisions, which agree to 230 digits (`qcos_analytic.py`). The sequence $|c_n| n^{3/2} |q_c|^n$ itself is strongly preasymptotic: at $n = 299$ it has reached only 0.78295, agreeing with the limit to three digits, so the coefficient asymptotics *corroborate* $R = |q_c|$ rather than establish it. What does establish $R = |q_c|$ is the fold itself, an actual singularity of the analytic continuation at that modulus. Neither q_c nor z_c satisfies an integer polynomial relation of degree ≤ 8 with coefficients up to 10^{14} , nor does the pair satisfy a bilinear one (PSLQ at 230 digits, tolerance 10^{-200}). Larger pair profiles are not reported: 230 digits do not support more than about 16 unknowns against a 10^{14} coefficient bound, and beyond that PSLQ returns vectors that re-evaluation at higher precision exposes as spurious.

(f) No change of variable rescues the ledger of (c): if $q = \psi(u)$ with $\psi \in u\mathbb{Z}[[u]]$ (as required to keep $z_1 \circ \psi \in \mathbb{Z}[[u]]$) and $u^* = \varphi(q^*) = s'/t' \in \mathbb{Q}$, the criterion again reads $|u^*| t' = s' < R_u$; but a non-polynomial integer power series has radius at most 1, so $R_u \leq 1$ while $s' \geq 1$. Every integral uniformization is capped at the boundary case $s' = 1 = R_u$: the sub-unit radius can be moved but not beaten. This is the geometric face of the same boundary on which the theta and Rogers–Ramanujan series sit in the truncation ledger (Remark 23).

(g) The cap has an elementary final form, the *discreteness cap*: for any non-polynomial $X \in \mathbb{Z}[[q]]$, any $q_0 = s/t$ inside its disc of convergence (away from the boundary), and any N , the truncation tail satisfies $|X(q_0) - X_N(q_0)| = |c_M|(s/t)^M(1 + o(1)) \geq s^M t^{-M}(1 - o(1))$, where $M > N$ is the first index with $c_M \neq 0$: at least the Liouville floor t^{-M} of the truncation's denominator whenever $s \geq 1$, because a nonzero integer coefficient has $|c_M| \geq 1$. (Measured exactly on z_1 at $q = \frac{1}{7}$: tail/floor = 1.533, 11.56, 70.86, 541.3 at $N = 6, 10, 14, 18$; `qcos_ledger.py`.) Integrality thus caps its own ledger: the property that grants linear heights also forbids the tail from undercutting the floor. (h) The arithmetic class of z_1 is bounded from above as well. No linear differential equation $\sum_{i \leq r} P_i(q) z_1^{(i)} = 0$ with $\deg P_i \leq d$ exists for $(r, d) \in \{(4, 10), (5, 12), (6, 14), (8, 10)\}$: the coefficient matrix, built from z_1 computed to order 4000, has full column rank (55, 78, 105 and 99 columns respectively) modulo each of two primes. A relation over $\mathbb{Q}(q)$ may be scaled to a primitive integer one, which survives reduction, so a full modular column rank does exclude a relation over $\mathbb{Q}$; what it cannot do is reach beyond the four profiles tested. The honest statement is therefore *not holonomic of order at most 8 within these degree windows*, and holonomy at higher order or degree remains open. Subject to that, z_1 is an integer power series that is non-algebraic and non-holonomic within the profiles tested and of radius < 1 , a class for which no arithmetic technology applies: the Chudnovsky–André–Bost theory and the arithmetic holonomy bounds of Calegari–Dimitrov–Tang

[3], which are built for “integrality plus structure”, all require holonomy, while Pólya–Carlson [23, 32] requires radius exactly 1. Integrality alone, without holonomy, is not currently actionable.

(i) One exact relation ties the lattice across the bases q , $-q$, q^2 : applying the zero product of Theorem 2 at the three bases and using $(q; q^2)_\infty(-q; q^2)_\infty = (q^2; q^4)_\infty$,

$$\prod_{k \geq 1} \varphi_k = 1, \quad \varphi_k := \frac{u_k(q) u_k(-q)}{u_k(q^2)}.$$

The relation is not termwise: $\varphi_1 = 1 + 2q^6 + 4q^8 + 8q^{10} + 18q^{12} + \dots$, $\varphi_2 = 1 - 2q^6 - \dots$, $\varphi_3 = 1 + 2q^{20} + \dots$, $\varphi_4 = 1 - 2q^{28} - \dots$ (order 300; adjacent leading defects cancel in pairs, and the truncated product over $k \leq 4$ is $1 + O(q^{54})$). No Mahler structure is found. With u_1 computed to order 4000 and every rank taken modulo each of two primes (a primitive integer relation survives reduction, so full column rank modulo one prime excludes a rational relation outright), u_1 satisfies no polynomial relation $P(q, u_1(q), u_1(q^d)) = 0$ for $d = 2, 3$ at bidegrees $(490, 1, 1)$, $(150, 2, 2)$, $(100, 3, 3)$, $(60, 4, 4)$, $(40, 5, 5)$, $(30, 6, 6)$, $(24, 7, 7)$, $(20, 8, 8)$; no order-two system relation $P(q, u_1(q), u_1(q^2), u_1(q^4)) = 0$ at $(30, 2, 2, 2)$, $(20, 3, 3, 3)$; no difference–differential mix $P(q, u_1, \theta u_1, u_1(q^2)) = 0$ at $(60, 2, 2, 2)$, $(40, 3, 3, 3)$, $(30, 4, 4, 4)$; no quasi-modular coupling, in the sense that the six generators $\{1, \theta \log u_1(q), \theta \log u_1(q^2), E_2(q), E_2(q^2), E_4(q)\}$ admit no $\mathbb{Q}(q)$ -linear relation with polynomial coefficients of degree up to 399 (2400 columns, full rank; so u_1 is not a hidden eta-quotient or E_2 -ring object of level 2 within that window); no linear relation among $\{1, \log u_1(q), \log u_1(q^2)\}$ to degree 665; and no Mahler relation for the logarithmic derivative $\theta \log u_1$ itself at $(150, 2, 2)$, $(100, 3, 3)$, $(60, 4, 4)$. All twenty-six profiles are regenerated by `tools/qcos_rank`, which prints the row count, the column count and the rank at each prime. Mahler’s method thus finds no foothold on the zero curve at any complexity reached here; as in (a) and (h), that is a statement about the profiles tested and not about all profiles.

(j) The fold is the natural point for a rationality reductio, because it is the unique point of the (q, z) -plane carrying *two* vanishings ($G = \theta G = 0$; a triple point $G = \theta G = \theta^2 G = 0$ is codimension three in two variables, hence absent). The first-jet web there consists of six values, $A = G(Qz_c)$, $B = \theta G(Qz_c)$, $h = H(z_c)$, $h_1 = H(Qz_c)$, $\eta = \theta H(z_c)$, $\eta_1 = \theta H(Qz_c)$, where H is the sine-side companion, which satisfies the mirror equation $H(z) - a(z)H(Qz) + qH(Q^2z) = 0$ (the $z^{1/2}$ in the second solution $z^{1/2}H$ moves the outer q). Exactly two finite-height relations hold: the Casoratian at z_c , $Ah = 1 - q_c$, and its θ -derivative, $Bh = -A\eta$ (both reproduced to 230 digits; `qcos_analytic.py`). For any mixed jet-Casoratian $C_{u,v}(z) = qu(z)v(Qz) - u(Qz)v(z)$ with $L(u) = f$, $\tilde{L}(v) = \tilde{g}$ the a -terms cancel in the increment:

$$C_{u,v}(Qz) - C_{u,v}(z) = \tilde{g}(z) u(Qz) - f(z) v(Qz),$$

so every telescoped sum of first-jet orbit bilinears closes back onto a Casoratian. In particular $B\eta = \sum_{m \geq 0} qQ^m z_c [G\theta H - H\theta G](Q^{m+1}z_c)$ (reproduced to 230 digits, 246 terms) is exact but carries an infinite rational series as coefficient: height-infinite. The ledger of a point evaluation is therefore fixed: each further θ - or δ -jet level adds at least three new values and exactly one finite-height relation, so the deficit (four at the fold, five at a generic zero) cannot be closed by jet-raising, and the fold’s two vanishings are the maximum the plane admits. Point evaluation alone, at any point including the fold, cannot complete a rationality reductio for this module.

(k) All of (c), (f), (g) are instances of a single elementary identity, which closes the truncation-ledger class outright.

Remark 19 (denominator–radius inequality; folklore). Let $F = \sum_{n \geq 0} f_n q^n$ with $f_n \in \mathbb{Q}$, not eventually zero; let $\mathcal{D}_n$ be the common denominator of $f_0, \dots, f_n$ and R the radius of convergence.

Then

$$R \leq \liminf_{n \in S} \mathcal{D}_n^{1/n}, \quad S := \{n : f_n \neq 0\}.$$

The restriction to S is necessary. Taking $f_n = c^{-n}$ when n is a power of 2 and $f_n = 0$ otherwise gives $R = c$ while $\liminf_n \mathcal{D}_n^{1/n} = c^{1/2}$, so the inequality fails over all n as soon as $c > 1$: the long runs of zeros hold $\mathcal{D}_n$ fixed while n grows. Only the integral case is used below, and it is unaffected. Consequently, if $F(s/t) = 0$ then the truncation ledger (which requires $s \mathcal{D}_N^{1/N} < R$) fails for every integer $s \geq 1$.

This is folklore, not a result of ours. For $\mathcal{D}_n \equiv 1$ it is the standard lemma that an integer power series of radius > 1 is a polynomial (Kedlaya [4, Lem. 11.1.2]; the surrounding circle of results is Pólya–Szegő [33, Part Eight, Ch. 3]), and the general case is the same argument after rescaling. Calegari, Dimitrov and Tang [3, Lem. 1.1.2] name the two halves the *Liouville lower bound* and the *Cauchy upper bound* and trace the lineage Borel (1894), Pólya [32] and Carlson [23], Dwork [25], Bertrandias, André, Bost; the hypothesis that the denominators grow at most geometrically is Christol’s *globally bounded* condition [24], and the G -function form of the corollary is in Rivoal [7, §3]. Note also that the inequality is generally *strict* ($f = \sum 2^n z^n$ has $\mathcal{D}_n \equiv 1$ and $R = \frac{1}{2}$), so it is not an identity.

Proof. Whenever $f_n \neq 0$ the number $\mathcal{D}_n f_n$ is a nonzero integer, so $|f_n| \geq 1/\mathcal{D}_n$ and hence $1/R = \limsup |f_n|^{1/n} \geq \limsup \mathcal{D}_n^{-1/n} = 1/\liminf \mathcal{D}_n^{1/n}$, which is the stated bound. The ledger criterion would then force $\mathcal{D}_N^{1/N} < R/s \leq R \leq \liminf \mathcal{D}_N^{1/N}$, a quantity strictly below its own lim inf. The bound is sharp: $f_n = c^{-n}$ has $\mathcal{D}_n = c^n$ and $R = c$ exactly. $\square$

The content is that *the denominator floor and the convergence radius are the same quantity*: buying radius by admitting denominators of growth c costs exactly c . Integral series are the case $\mathcal{D}_n = 1$, recovering (f) and (g); series with geometric denominators gain radius and lose precisely as much. This is why no rearrangement, reparametrization, or basis change in this paper’s ledgers has ever produced excess, and why none can.

A sharp illustration is available. The radius of z_1 is capped by the fold of item (e), where z_1 and z_2 collide; but a fold is a square-root branch point of each branch *separately* and not of the pair, so the elementary symmetric functions of the first m branches (integral after clearing the poles of $z_k = u_k q^{2(1-k)}$) are analytic through every collision among them, and the crossing condition survives as the statement that $2q(1-q)$ is a root of $X^m - e_1 X^{m-1} + \dots$. Their radii increase with m . Estimated from coefficient growth at order 300, by one and the same estimator for every m so that the rows are comparable, they are 0.558, 0.710, 0.778, 0.794 for $m = 1, 2, 3, 4$, improving the ledger factor $1/R$ from 1.79 to 1.26 (`qcos_ledger.py`). The $m = 1$ estimate is to be read against the exact value $|q_c| = 0.5546$ from item (e): the estimator reads about 0.5% high there, and the remaining three rows carry a comparable error, so the improvement is real but the individual figures are estimates and not certified radii. By Proposition 19 the whole family is capped at 1. The $m \rightarrow \infty$ limit is the zero product of Theorem 2, of radius exactly 1: the regularized product is the critical endpoint of this monotone family, which is the precise reason its arithmetic is confined to the tail.

Together, (c), (f), (g) and Remark 23 express one law from four sides: for every representation of this problem yet constructed, the arithmetic cost equals the analytic decay *exactly*, with zero excess; a proof of Conjecture 3 must produce strict excess, and rearrangement, reparametrization, or symmetrization of the object’s own $\mathbb{Q}(q)$ -structure provably cannot.

5.6 The second-kind forms and the exact reduction

The q -Siegel construction for this class is explicit. For $N \geq 1$ put

$$P_N(q, z) := (q; q)_{2N} \sum_{k=0}^N \frac{(-1)^k q^{k(k-1)} z^k}{(q; q)_{2k}} = \sum_{k=0}^N (-1)^k q^{k(k-1)} z^k \prod_{i=2k+1}^{2N} (1 - q^i) \in \mathbb{Z}[q, z].$$

Proposition 20 (the second-kind forms). $\deg_q P_N = 2N^2 + N$ and $\deg_z P_N = N$ (exact in $\mathbb{Z}[q, z]$; verified for $N \leq 8$ by `qcos_ledger.py`), and the remainder $R_N(q, z) := (q; q)_{2N} G(q, z) - P_N(q, z)$ satisfies, uniformly for q in compacts of $(0, 1)$ and z bounded,

$$|R_N(q, z)| = q^{N^2(1+o(1))}.$$

Proof. Membership in $\mathbb{Z}[q, z]$ and $\deg_z P_N = N$ are clear from the displayed product form, since $(q; q)_{2N}/(q; q)_{2k} = \prod_{i=2k+1}^{2N} (1 - q^i) \in \mathbb{Z}[q]$. For the q -degree, the k -th summand has degree $k(k-1) + \sum_{i=2k+1}^{2N} i = 2N^2 + N - k^2 - 2k$, maximal at $k = 0$ with value $\deg(q; q)_{2N} = 2N^2 + N$; the maximum is attained once, so no cancellation occurs. For the remainder, $R_N = (q; q)_{2N} \sum_{k>N} (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$, whose leading term ($k = N+1$) is $\pm q^{N(N+1)} z^{N+1} (q; q)_{2N} / (q; q)_{2N+2}$; the tail is dominated by a geometric series in q^{2N} , so $|R_N| = q^{N^2+O(N)}$ uniformly. (Measured at $q = 0.3$, the ratio $\log |R_N| / (N^2 \log q)$ falls to 1.088 at $N = 20$ on the diagonal $z = 2q(1 - q)$ and to 1.036 at a generic z_0 ; `qcos_ledger.py`.) $\square$

Proposition 21 (reduction to the Siegel margin). Let $q = s/t \in (0, 1)$ and let $z_0 = a/b \in \mathbb{Q}^\times$ with $G(q, z_0) = 0$. Then $P_N(q, z_0) = -R_N(q, z_0)$ and $t^{2N^2+N} b^N P_N(q, z_0) \in \mathbb{Z}$. Consequently, if for infinitely many N one has $P_N(q, z_0) \neq 0$ and

$$|R_N(q, z_0)| < t^{-(2N^2+N)} b^{-N},$$

then $G(q, z_0) \neq 0$: a contradiction. By Proposition 20 the displayed criterion reads $q^{N^2} < t^{-2N^2}$ up to $O(N)$ in the exponent, i.e.

$$\log(t/s) > 2 \log t \iff st < 1,$$

which is false. The deficit is exactly the factor 2 in $\deg_q P_N = \deg(q; q)_{2N}$: the double Pochhammer.

Proof. $P_N(q, z_0) = (q; q)_{2N} G(q, z_0) - R_N(q, z_0) = -R_N(q, z_0)$ by hypothesis; clearing denominators of $P_N \in \mathbb{Z}[q, z]$ at $q = s/t$, $z = a/b$ gives $t^{\deg_q P_N} b^{\deg_z P_N} P_N(q, z_0) \in \mathbb{Z}$, and a nonzero integer has modulus ≥ 1 . The stated evaluation of the criterion is Proposition 20. Every step of this proof, the clearing included, is machine-checked (§7). $\square$

Corollary 22 (what would suffice). Conjecture 3 at z_0 (hence, on the diagonal, the irrationality of β_2) follows from any family of forms $\Lambda_N \in \mathbb{Z}[q, z]$ with $\Lambda_N(q, z_0) \neq 0$, remainder $q^{N^2(1+o(1))}$ against G , and

$$\deg_q \Lambda_N \leq N^2 \frac{\log(t/s)}{\log t} (1 - \varepsilon), \quad \text{in particular } \deg_q \Lambda_N \leq N^2(1 - \varepsilon) \text{ when } s = 1.$$

The forms of Proposition 20 have $\deg_q = 2N^2$: it suffices to halve the q -degree at the same remainder, or equivalently to double the remainder decay to q^{2N^2} . A single-Pochhammer clearing $(q; q)_N$ would have degree $\frac{1}{2}N^2(1+o(1))$ and would suffice at once; no such representation of G along a non- q -power argument exists in the Rogers–Ramanujan/Bailey framework.

Remark 23 (the theta-decay barrier: why Corollary 22 is unreachable by truncation). The hypothesis of Corollary 22 cannot be met by *any* truncate-and-clear form. Let $\sum_k c_k(q) q^{\alpha k^2} z^k$ ($\alpha > 0$, $c_k \in \mathbb{Q}(q)$) be truncated at N and cleared. The $k = N$ summand already contributes $q^{\alpha N^2}$, so $\deg_q \geq \alpha N^2$; the remainder is the first omitted term, of size $q^{\alpha N^2(1+o(1))}$. The criterion $|R_N| < t^{-\deg_q}$ therefore reads

$$\alpha N^2 \log(t/s) > \alpha N^2 \log t \iff s < 1,$$

false for every $q = s/t \in (0, 1)$: *the decay and the degree are driven by the same factor $q^{\alpha k^2}$ and cancel exactly*. Denominators only make matters worse. The cleared truncations have $\deg_q = N^2$ exactly for the theta series $\sum q^{k^2} z^k$ and for the Rogers–Ramanujan series $\sum q^{k^2}/(q; q)_k$, so both sit at $\beta := \deg_q/N^2 = 1$, the boundary; the Hahn–Exton series sits at $\beta = 2 + 1/N$, that is at 2.25, 2.17, 2.12 for $N = 4, 6, 8$, a further factor 2 past a boundary that is itself already unreachable (`qcos_ledger.py`).

This describes the shape of the known landscape and not merely our position in it. The transcendence of theta values and of Rogers–Ramanujan values, both at $\beta = 1$, is known *only* through Nesterenko’s modular method [22] and never through a Siegel/Liouville margin, precisely because no theta-decay series can cross $\beta < 1$. The two mechanisms left for Conjecture 3 are therefore modularity and a multiplicity theory for the associated (σ, δ) -module. For the first, no analogue of [22] is known off the modular class, and we have not found G inside it; that is the state of knowledge, not a proof that G lies outside (see [14]). The double Pochhammer is not the whole obstruction: it is a factor 2 on top of a structural one.

5.7 An unconditional effective bound

Although Conjecture 3 is open, the specific root q^* admits an effective non-rationality statement, proved by certified interval arithmetic.

Proposition 24 (effective non-rationality of q^*). *The least positive root q^* of S admits no rational representation s/t with $t \leq 10^{1044}$. Consequently, if $\beta_2 = u/v$ in lowest terms then $u > 10^{522}$. The method is not new: it is Brent’s [2], who proved by the same continued-fraction argument that a rational representation of Euler’s constant needs a large denominator, and it rests on Khinchin’s Theorem 17 (every convergent is a best approximation of the second kind) [31, Thm. 17]. Lagarias [5, §3.15] surveys the Brent–McMillan denominator bounds for Euler’s constant and quotes Khinchin’s theorem in their proof. What is new here is only the bound for this particular constant.*

Proof. All steps are certified interval arithmetic (outward rounding; `mpmath.iv`; script `beta2_effective_irrational` which bootstraps q^* from a 25-digit start by Newton at doubling precision and so needs no stored input), with the series tail bounded by $(q; q)_{2k} \geq (1 - 2q)/(1 - q) > 0$ for $q < \frac{1}{2}$ and term ratios $< \frac{1}{2}$ beyond the truncation. (i) $S > 0$ on $[0, 0.40]$ (400 certified chunks). (ii) $S' \leq -1$ on $[0.40, 0.46]$ (600 certified chunks; S' evaluated term-by-term with the logarithmic-derivative weights), so S has at most one zero in $[0.40, 0.46]$. (iii) with c the 2090-digit truncation of q^* , $a = c - 2 \cdot 10^{-2090}$ and $b = c + 2 \cdot 10^{-2090}$, certified evaluation gives $S(a) > 0 > S(b)$ (working precision 2130 digits, series tail enclosed in $\pm 10^{-2125}$); by (i)–(ii), $q^* \in [a, b]$. (iv) the interval $[a, b]$ contains the rational s_0/t_0 (Stern–Brocot construction) with $9.5 \cdot 10^{1044} < t_0 < 10^{1045}$, verified by exact rational comparison, and certified $S(s_0/t_0) \neq 0$. If $m/n \in [a, b]$ with $n \leq 10^{1044}$ and $m/n \neq s_0/t_0$, then $|m/n - s_0/t_0| \geq 1/(nt_0) > 10^{-2089} > b - a = 4 \cdot 10^{-2090}$, impossible; so $m/n = s_0/t_0$, whence $n = t_0 > 10^{1044}$, a contradiction. Thus no rational with denominator $\leq 10^{1044}$ lies in $[a, b]$ at all, in particular none equals q^* . For $\beta_2 = u/v$ in lowest terms, $q^* = v^2/u^2$ in lowest terms, so $u^2 > 10^{1044}$. The Diophantine half of this argument, steps (iv) onward, is machine-checked, with steps (i)–(iii) as hypotheses (§7). $\square$

6 Atom decomposition and status

The table below is the paper’s status register. Every numbered statement, and every remark carrying mathematical content, appears exactly once, with a one-line paraphrase, its label, the statements it depends on, and its status. PROVED means a complete human-style proof is written out here or cited from the literature; HEURISTIC means the support is computational; CONJECTURE means believed and unproved. A statement is exactly as strong as the weakest label in its dependency chain, and the table is arranged so that this can be checked by reading upwards. A star marks a statement at PROVED whose formalization is incomplete; §7 records, for each starred entry, the coverage attained and what blocks the rest. There are no unlabelled mathematical statements in the paper and no atom without a path to Conjecture 3 or to Proposition 24.

ID	statement	label	depends on	status
A1	Thm 1: $G(q, \cdot)$ satisfies no algebraic differential equation over $\mathbb{C}(z)$	PROVED	[10]	done*, known
A2	Lem 4: the rank-2 σ -module is irreducible; the Riccati equation has no rational solution	PROVED	none	done*
A3	Thm 5: the difference Galois group contains SL_2	PROVED	A2, [34]	done*
A4	Prop 6: no $\sigma\delta$ -integrability relation, not even with a scalar twist	PROVED	none	done*
A5	Cor 7: $G(q, \cdot)$ is differentially transcendental, by the parametrized route	PROVED	A3, A4, [15]	done*
A6	Thm 8 (= Thm 2): $C_1 = 1/(q; q)_\infty$ and $\prod_k z_k Q^{k-1} = (q; q^2)_\infty$	PROVED	none	done*
A7	Rem 9: if z^* is algebraic then the tail product is transcendental	PROVED	A6, [22]	done*
A8	Thm 10: the Hensel root u_k of F_k lies in $1 + q\mathbb{Z}[[q]]$	PROVED	A6	done*; the identification of u_k with $q^{2(k-1)}z_k$ needs convergence, hence A22
A9	Lem 11: the parameter–argument swap for ${}_1\phi_1(0; b; Q, z)$	PROVED	[18]	done*, known
A10	Prop 12: $\mathrm{ord}_q G(q, Q^{-m}) = (m+1)^2$ with leading coefficient $(-1)^{m+1}$	PROVED	A9, [18]	done*, known
A11	Cor 13: $\mathrm{onset}(u_k - 1) = k(2k-1)$, leading coefficient -1	PROVED	A8, A10	done*
A12	Rem 14: order reflection $-\frac{1}{2} \mapsto m+1$; backward-orbit constant; vanishing transferred to the parameter slot	PROVED	A9, A10	done*
A13	Prop 15: the sine-side integrality, lattice values and onsets $k(2k+1)$	PROVED	A8, A9	done*
A14	Rem 16: the two onset families exhaust the triangular numbers, each once	PROVED	A11, A13	done*
A15	Prop 17: both deviation families stabilize q -adically to $1/(q^2; q^2)_\infty^2$	PROVED	A10, A11, A13	done*
A16	Rem 19: $R \leq \liminf_{n \in S} \mathcal{D}_n^{1/n}$, and the truncation ledger fails for every integer $s \geq 1$	PROVED	none	done*, folklore
A17	Prop 20: $\deg_q P_N = 2N^2 + N$, $\deg_z P_N = N$, $ R_N = q^{N^2(1+o(1))}$	PROVED	none	done*
A18	Prop 21: the reduction to the Siegel margin, and its failure by the factor 2	PROVED	A17	done*
A19	Cor 22: halving the q -degree at the same remainder would suffice	PROVED	A18	done*
A20	Rem 23: no truncate-and-clear form meets A19's hypothesis	PROVED	A19	done*
A21	Rem 18, items (c), (f), (g), (j), (k): the linear-height ledger, the uniformization cap, the discreteness cap, the jet deficit at the fold, and their common source	PROVED	A8, A16	done*
A22	Rem 18, items (b), (d), (e): the radius equals $ q_c $; the fold coordinates and the fold constant; the coefficient sequence is new	HEURISTIC	A8	numerical, 230 digits
A23	Rem 18, items (a), (h), (i): the exclusions of algebraic, holonomic, Mahler, system, quasi-modular and log-linear relations	PROVED within the profiles listed, HEURISTIC beyond them	A8	rank certificates
A24	Prop 24: q^* has no rational representation of denominator $\leq 10^{1044}$	PROVED	A25	done*
A25	The certified enclosure of q^* and the Stern–Brocot fraction s_0/t_0	PROVED	none	interval arithmetic; hypothesis of A24
A26	Conj 3: every positive zero $z_k(q)$ is transcendental at algebraic q	CONJECTURE	none	open; the goal

Reading of the table. A26 is the goal; A1–A5 place the function outside the reach of differential-algebraic methods, A6–A15 supply the exact structure of the zero lattice, and A16–A24 map the negative space: which mechanisms cannot reach A26 and what the best unconditional statement about the distinguished zero currently is. A22 and A23 are the only entries not at `PROVED`, and neither is used in the proof of any other entry. A8 is stated for the formal Hensel root and is unconditional as such; only its identification with the analytic k -th zero passes through A22, and that qualification is carried in the theorem itself rather than in a remark. A23 is terminal. A25 is separated from A24 because it is exactly the part of Proposition 24 that a machine check consumes as a hypothesis rather than proves; §7 says so as well.

7 Machine verification

The arithmetic content of this paper is formalised in Lean 4 (toolchain v4.30.0, Mathlib v4.30.0). Five files in `lean/with_mathlib/` of the cited repository carry it:

`QCosineExponents`, `QCosineLattice`, `QSiegelLedge`, `QEffectiveExclusion`, `QZeroSeries`.

All five are registered in `lakefile.toml` and build with a bare `lake build`. Every declaration named below is `sorry`-free, and its axiom list was read off a cold build with `#print axioms`. The only axioms occurring are Lean’s standard `propext`, `Classical.choice` and `Quot.sound`, with the single exception of `QZeroSeries.lean`, where each theorem is decided by compiled evaluation and therefore also carries the reflection axiom that `native_decide` generates.

Statement	Coverage	Lean declarations
Thm 1	none	(see below)
Thm 2	exponent steps	as Thm 8
Lem 4	none	(see below)
Thm 5	exponent gap	supermult_exponent, gap_ge, gap_eq_at_ends
Prop 6	none	(see below)
Cor 7	none	(see below)
Thm 8	exponent steps	zeroproduct_reindex, zeroproduct_peak_bound, zeroproduct_peak_slack, euler_split
Thm 10	inputs, and order q^{80}	integrality_exponent, newtonExp_nonneg, newtonExp_eq_zero_iff, newtonExp_second_difference, newton_residual_shape, residualPoly_eval_one, residualPoly_derivative_eval_one, residualPoly_derivative_isUnit, qPochhammer_isUnit, u1_coefficients, u2_prefix, cosine_residual_vanishes
Lem 11	exponent identities	swap_symmetry_exponent, swap_base_collapse
Prop 12	no-cancellation step	lattice_order_gap, lattice_order_strictMono, lattice_order_leading
Cor 13	onsets, $k \leq 6$	hexagonal_onset, hexagonal_next_correction, cosine_onset, cosine_onsets
Prop 15	onsets, $k \leq 5$	sine_shift, sine_onset, onsets_differ, onsets_consecutive, sine_onsets, sine_residual_vanishes
Prop 17	depths, $k \leq 6$	stablelaw_numerator_depth, stablelaw_numerator_depth_sharp, stablelaw_pair_exponent, stablelaw_pair_weight, stablelaw_pair_parity, stablelaw_denominator_depth, twoColoured_prefix, cosine_agreement_depth, sine_agreement_depth
Rem 19	Liouville half	one_le_abs_of_ne_zero
Prop 20	degrees	tail_sum, degree_ledger, degree_at_zero, degree_lt_of_pos
Prop 21	complete	prod_one_sub_div, clearedTerm_eq, cleared_eq, reduction, margin_iff, margin_fails, st_ge_one
Cor 22	degree threshold at the coefficient 2	margin_iff (see below)
Rem 23	complete	theta_barrier
Prop 24	complete	farey_gap, unique_rational_in_interval, no_rational_of_denominator_le, denominator_gt_of_mem, width_lt_gap, qstar_no_small_denominator

Two entries deserve comment because the formalisation changed the text. The displayed derivation of Proposition 15(c) previously omitted the prefactor $q^{k(k-1)}$ and so evaluated to 1 rather than to $k(2k+1)$; and the numerator step of Proposition 17 previously gave the relative order of the r -th term as $(k+r)^2 - k(2k-1)$, which at $k=2, r=1$ is 3 and contradicts the depth $2k+1=5$ asserted two lines later. Both are corrected above; neither conclusion changes.

The following are *not* formalised.

- *All of §3.* Theorem 5, Proposition 6, Corollary 7 and Theorem 1 are statements of difference Galois theory and of its parametrized refinement. Mathlib contains no difference Galois theory, no q -difference modules and no Picard–Vessiot theory for difference equations, so there is nothing to reduce them to; formalising them means formalising [16] and [15] first. Only the exponent gap that replaced the numerical check in the proof of Theorem 5 is machine-checked. A file whose content was “assume [15, Prop. 2.9], conclude” would add nothing and is deliberately not supplied. The same applies to Lemma 4, whose argument is a valuation-theoretic exhaustion over the Q -orbits of $\mathbb{C}^*$.
- *The analytic steps of Theorem 8:* the uniform peak estimate, the Rouché localisation and the dominated-convergence passage. The exponent identities those steps consume are formalised;

the estimates are not. Mathlib has the dominated convergence theorem but no Rouché theorem, so step (3) has no target to reduce to at present.

- *The Hensel step of Theorem 10.* Mathlib does supply `IsAdicComplete` (`Ideal.span {X}`) (`PowerSeries R`) together with `IsAdicComplete.henselianRing`, so $\mathbb{Z}[[q]]$ is Henselian at (q) in the library's sense. That instance does not apply here: `HenselianRing.is_henselian` requires a *monic polynomial*, whereas F_k is a power series in u whose top behaviour is not a unit. The missing ingredient is a Weierstrass-preparation form of the implicit function theorem for $A[[u]]$ over an adically complete A that is not local, and Mathlib's Weierstrass preparation assumes A local. What is verified instead is the finite-order content, which is the part the paper displays: the Newton recursion carried out entirely over $\mathbb{Z}$, with the residual $F_k(q, u_k)$ checked to vanish to order q^{80} for $k \leq 6$ on the cosine side and $k \leq 5$ on the sine side.
- *The Cauchy half of Remark 19*, namely $1/R = \limsup |f_n|^{1/n}$. The Liouville half, $|f_n| \geq 1/\mathcal{D}_n$ on the support, is formalised; the radius identity is Cauchy–Hadamard and is used as such.
- *The remainder estimate of Proposition 20*, $|R_N| = q^{N^2(1+o(1))}$. It is an estimate on a q -series tail and enters `reduction` as a hypothesis, which is the honest place for it: what is proved is that the estimate, together with $P_N(q, z_0) \neq 0$, forces a contradiction.
- *The interval arithmetic of Proposition 24:* the enclosure of q^* , the Stern–Brocot rational s_0/t_0 and the certified sign changes come from `beta2_effective_irrationality.py` and enter `qstar_no_small_denominator` as hypotheses. The Diophantine argument that consumes them, including the width comparison $4 \cdot 10^{-2090} < 1/(10^{1044}t_0)$ for every $t_0 < 10^{1045}$, is formalised in full.
- *The general degree threshold of Corollary 22.* Its content is the comparison already formalised as `margin_iff`, which decides $2 \log t < \log(t/s) \iff st < 1$: that is the threshold at the coefficient 2 realised by Proposition 20. What is not formalised is the same comparison with the coefficient 2 replaced by the parameter $\deg_q \Lambda_N/N^2$, together with the $o(1)$ bookkeeping in the remainder hypothesis. No Mathlib gap blocks this: the threshold is one inequality between logarithms and the asymptotic hypothesis is expressible with `Filter.Tendsto`. It is open formalization work, and it is recorded here as such rather than left without a status.
- *The exclusion searches of Remark 18*, which are rank computations on large integer matrices over $\mathbb{Z}/p$. Each is a certified rank witness and nothing in Mathlib blocks formalising it; what would be needed is to import the coefficient data and the elimination transcript, which is a data-volume question and not a mathematical one. They are reported as computations, regenerated by `tools/qcos_rank`.

One point of method is worth recording, since it is what made Proposition 21 cheap to verify. Denominator clearing is normally stated for a general two-variable integer polynomial, in terms of its two total degrees, and proving it in that form is substantial. It is not needed. Since P_N is given by an explicit product formula, the cleared quantity can be written down as an integer,

$$t^{2N^2+N} b^N P_N(s/t, a/b) = \sum_{k=0}^N (-1)^k s^{k(k-1)} a^k b^{N-k} t^{k^2+2k} \prod_{i=2k+1}^{2N} (t^i - s^i),$$

and the whole content becomes the exponent ledger $k(k-1) + \sum_{i=2k+1}^{2N} i + (k^2 + 2k) = 2N^2 + N$, which is `degree_ledger`.

Declarations

Funding. No funding was received for this work.

Competing interests. The author declares no competing interests.

Data availability. All code, numerical certification data and Lean sources supporting the computational and machine-checked statements are openly available in the repository cited on the title page and archived on Zenodo (concept DOI 10.5281/zenodo.20370090). The Lean files are listed in §7.

Use of AI tools. Large language models (Anthropic Claude) were used in the preparation of this manuscript, both in drafting and in computation. Every computational claim is certified by the scripts in the cited repository, independently of any model output. The author takes full responsibility for the content.

References

- [1] M. Amou, M. Katsurada, K. Väänänen, *On the values of certain q -hypergeometric series II*, in: Analytic Number Theory, eds. C. Jia, K. Matsumoto, Developments in Mathematics **6**, Kluwer Acad. Publ., Boston, MA, 2002, 17–25.
- [2] R. P. Brent, *Computation of the regular continued fraction for Euler’s constant*, Math. Comp. **31** (1977), 771–777.
- [3] F. Calegari, V. Dimitrov, Y. Tang, *The unbounded denominators conjecture*, J. Amer. Math. Soc. **38** (2025), no. 3, 627–702; arXiv:2109.09040.
- [4] K. S. Kedlaya, *Weil cohomology in practice*, lecture notes, <https://kskedlaya.org/weil-cohom/>, Ch. 11 (Dwork’s proof of rationality), Lemma 11.1.2. These are living notes with no version stamp; the statement used is that an integer power series with radius of convergence > 1 is a polynomial, whose two-line proof is reproduced above.
- [5] J. C. Lagarias, *Euler’s constant: Euler’s work and modern developments*, Bull. Amer. Math. Soc. **50** (2013), 527–628.
- [6] L. Lorch, M. E. Muldoon, *Transcendentality of zeros of higher derivatives of functions involving Bessel functions*, Int. J. Math. Math. Sci. **18** (1995), 551–560.
- [7] T. Rivoal, *Les E -fonctions et G -fonctions de Siegel*, in: Journées mathématiques X-UPS, Année 2019: Périodes et transcendance, Éd. P. Harinck, A. Plagne, C. Sabbah, Les Éditions de l’École polytechnique / Centre Mersenne, 2019, 197–298.
- [8] F. Lindemann, *Ueber die Zahl π* , Math. Ann. **20** (1882), 213–225.
- [9] C. L. Siegel, *Über einige Anwendungen diophantischer Approximationen*, Abh. Preuss. Akad. Wiss., Phys.-math. Kl. (1929), Nr. 1, 70 pp.; Gesammelte Abhandlungen I, Springer, 1966, 209–266. English translation by C. Fuchs, ed. U. Zannier, Quaderni della Scuola Normale Superiore di Pisa, Monographs 2, Edizioni della Normale, 2014, 81–138.
- [10] B. Adamczewski, T. Dreyfus, C. Hardouin, *Hypertranscendence and linear difference equations*, J. Amer. Math. Soc. **34** (2021), 475–503.

- [11] L. D. Abreu, J. Bustoz, J. L. Cardoso, *The roots of the third Jackson q -Bessel function*, Int. J. Math. Math. Sci. **2003**, no. 67, 4241–4248.
- [12] M. Amou, T. Matala-aho, K. Väänänen, *On Siegel–Shidlovskii’s theory for q -difference equations*, Acta Arith. **127** (2007), 309–335.
- [13] C. E. Arreche, T. Dreyfus, J. Roques, *Differential transcendence criteria for second-order linear difference equations and elliptic hypergeometric functions*, J. Éc. polytech. Math. **8** (2021), 147–168.
- [14] V. Bonfiori, *Transcendence of a planar reflection–group growth series and its relaxed companion*, preprint (2026), archived at doi:10.5281/zenodo.20370090 (file paper/journal/paper2.pdf).
- [15] T. Dreyfus, C. Hardouin, J. Roques, *Hypertranscendence of solutions of Mahler equations*, J. Eur. Math. Soc. **20** (2018), 2209–2238.
- [16] C. Hardouin, M. F. Singer, *Differential Galois theory of linear difference equations*, Math. Ann. **342** (2008), 333–377.
- [17] M. E. H. Ismail, C. Zhang, *Zeros of entire functions and a problem of Ramanujan*, Adv. Math. **209** (2007), 363–380.
- [18] T. H. Koornwinder, R. F. Swarttouw, *On q -analogues of the Fourier and Hankel transforms*, Trans. Amer. Math. Soc. **333** (1992), 445–461.
- [19] T. Morita, *A connection formula of the Hahn–Exton q -Bessel function*, SIGMA **7** (2011), 115, 11 pp.
- [20] S. Nishioka, *Proof of unsolvability of q -Bessel equation using valuations*, J. Math. Sci. Univ. Tokyo **23** (2016), 763–789.
- [21] H. T. Koelink, R. F. Swarttouw, *On the zeros of the Hahn–Exton q -Bessel function and associated q -Lommel polynomials*, J. Math. Anal. Appl. **186** (1994), 690–710.
- [22] Yu. V. Nesterenko, *Modular functions and transcendence questions*, Sb. Math. **187** (1996), 1319–1348.
- [23] F. Carlson, *Über Potenzreihen mit ganzzahligen Koeffizienten*, Math. Z. **9** (1921), 1–13.
- [24] G. Christol, *Globally bounded solutions of differential equations*, in: Analytic Number Theory (Tokyo, 1988), Lecture Notes in Math. **1434**, Springer, Berlin, 1990, 45–64.
- [25] B. Dwork, *On the rationality of the zeta function of an algebraic variety*, Amer. J. Math. **82** (1960), 631–648.
- [26] H. Exton, *A basic analogue of the Bessel–Clifford equation*, Jñānābha **8** (1978), 49–56.
- [27] G. Gasper, M. Rahman, *Basic Hypergeometric Series*, 2nd ed., Encyclopedia of Mathematics and its Applications **96**, Cambridge University Press, Cambridge, 2004.
- [28] W. Hahn, *Beiträge zur Theorie der Heineschen Reihen. Die 24 Integrale der hypergeometrischen q -Differenzgleichung. Das q -Analogon der Laplace-Transformation*, Math. Nachr. **2** (1949), no. 6, 340–379.

- [29] W. K. Hayman, *A generalisation of Stirling's formula*, J. Reine Angew. Math. **196** (1956), 67–95.
- [30] C. G. J. Jacobi, *Fundamenta Nova Theoriae Functionum Ellipticarum*, Sumtibus fratrum Bornträger, Regiomonti, 1829; Gesammelte Werke I, Reimer, Berlin, 1881.
- [31] A. Ya. Khinchin, *Continued Fractions*, transl. Scripta Technica, ed. H. Eagle, University of Chicago Press, Chicago, 1964; reprinted Dover, Mineola NY, 1997.
- [32] G. Pólya, *Über Potenzreihen mit ganzzahligen Koeffizienten*, Math. Ann. **77** (1916), 497–513.
- [33] G. Pólya, G. Szegő, *Problems and Theorems in Analysis II*, transl. C. E. Billigheimer, Springer, Berlin, 1976; reprinted in Classics in Mathematics, 1998.
- [34] J.-P. Ramis, J. Sauloy, C. Zhang, *Local analytic classification of q -difference equations*, Astérisque **355**, Soc. Math. France, 2013.
- [35] M. van der Put, M. F. Singer, *Galois Theory of Difference Equations*, Lecture Notes in Mathematics **1666**, Springer, Berlin, 1997.